

Linear Regression in R: OLS Assumptions and the Boston Housing Data

Regression Families

A focused guide to **linear regression** using the `daltoolbox` package in R — covering both the theoretical foundations and practical implementation with the Boston dataset. The experimental workflow remains consistent throughout, making it easy to follow from simple baselines to multiple predictors.

01

Linear Regression

The baseline method that assumes a linear relationship between predictors and the target variable — in both simple (one predictor) and multiple (several predictors) forms. Interpretable, efficient, and essential as a reference point for evaluating more complex models.

The Boston Dataset

All examples in this series use the `Boston` dataset from the `MASS` package. The target variable is **medv** (median house value). A consistent train/test split is applied across all models via `set.seed(1)`, ensuring fair and comparable evaluations.

Standard Setup

```
library(daltoolbox)
library(MASS)
data(Boston)
Boston <- as.matrix(Boston)
set.seed(1)
sr <- sample_random()
sr <- train_test(sr, Boston)
boston_train <- sr$train
boston_test <- sr$test
```

Key Variables

- **crim** — per capita crime rate
- **rm** — average number of rooms per dwelling
- **lstat** — % of lower-income population
- **nox** — nitric oxide concentration
- **medv** — median house value (*target variable*)

All features are numeric or integer. Converting to a matrix is optional, but may improve performance for certain methods.

Linear Regression

Linear regression models the relationship between a **dependent variable** Y and one or more **independent variables** X through a linear equation of the form $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$. The coefficients β are estimated via the **Ordinary Least Squares (OLS)** estimator, which minimizes the sum of squared residuals between observed and fitted values — yielding the best linear unbiased estimates under standard conditions.

Valid inference from OLS depends on four core assumptions:

Linearity

The relationship between predictors and response is linear in parameters.

Independence

Observations and error terms are independent of one another.

Homoscedasticity

The variance of residuals is constant across all levels of the predictors.

Normality

Residuals are approximately normally distributed, enabling valid hypothesis tests.

📖 James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer. Chapter 3.

Linear Regression (reg_lm)

reg_lm implements **Ordinary Least Squares (OLS) multiple linear regression**, estimating one coefficient per predictor plus an intercept. The prediction is a **weighted linear combination** of the input features — making the model highly interpretable, as each coefficient directly quantifies the marginal effect of its corresponding predictor on the target variable.

The model predicts the target $\hat{y}$ according to the following equation, where each β_i is estimated by minimizing the **Residual Sum of Squares (RSS)** — the sum of squared differences between observed and fitted values:

$$\hat{y} = \beta_0 + \beta_1x_1 + \beta_2x_2 + \cdots + \beta_px_p$$

Model performance is assessed using three complementary metrics:

MSE

Mean Squared Error — average squared deviation between predicted and actual values. Lower is better; penalizes large errors heavily.

SMAPE

Symmetric Mean Absolute Percentage Error — scale-independent relative error. Lower is better; treats over- and under-prediction symmetrically.

R²

Coefficient of Determination — proportion of variance in the target explained by the model. Ranges from 0 to 1; higher is better.

Training Code

```
model <- reg_lm("medv")
model <- fit(model, boston_train)
```

Training Metrics

MSE: 21.74 | SMAPE: 0.17 | R²: 0.75

Test Metrics

MSE: 25.31 | SMAPE: 0.19 | R²: 0.67

❏ **Didactic goal:** Linear regression is the ideal baseline model — each coefficient reveals the **direction** (positive or negative) and **magnitude** of each predictor's effect on the target variable `medv`. Simple, interpretable, and effective for data with linear trends, it establishes the benchmark against which all subsequent models will be compared.

❏ Montgomery, D. C., Peck, E. A., & Vining, G. G. (2012). *Introduction to Linear Regression Analysis* (5th ed.). Wiley. — Classic reference for OLS theory and diagnostics.

reg_lm: Linear Regression with daltoolbox

Setup & Data

```
# Linear Regression

#install.packages("daltoolbox")
library(daltoolbox)
library(MASS)

data(Boston)
Boston <- as.matrix(Boston)

set.seed(1)
sr <- sample_random()
sr <- train_test(sr, Boston)
boston_train <- sr$train
boston_test <- sr$test
```

Training & Evaluation

```
# Training
model <- reg_lm("medv")
model <- fit(model, boston_train)

# Evaluation
train_pred <- predict(model, boston_train)
train_eval <- evaluate(model,
  boston_train["medv"], train_pred)
print(train_eval$metrics)
# mse=21.74, smape=0.17, R2=0.75

test_pred <- predict(model, boston_test)
test_eval <- evaluate(model,
  boston_test["medv"], test_pred)
print(test_eval$metrics)
# mse=25.31, smape=0.19, R2=0.67
```

Step-by-Step

- Load `daltoolbox` and the `Boston` dataset (MASS)
- Convert to matrix with `as.matrix()` for framework compatibility
- Split train/test using `sample_random()` and `train_test()`
- Define the model with `reg_lm("medv")` and fit with `fit()`
- Evaluate with `evaluate()`: metrics MSE, SMAPE, and R^2
- Compare train ($R^2=0.75$) vs. test ($R^2=0.67$) — a small gap indicates good generalization

❏ **Key concept:** Linear regression assumes a linear relationship between predictors and the target variable. Each coefficient indicates the **direction** (positive or negative) and **magnitude** of each predictor's effect on `medv`. It serves as the ideal baseline model: simple, interpretable, and efficient for data with linear trends — setting the foundation for the more complex methods that follow.

Acknowledgements



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